

One Dimensional Cellular Automata: Application of Evolutionary Systems for Sea Level Analysis

Raquel Silva dos Santos^{1,*}

¹Department of Statistics and Informatics Federal Rural University of Pernambuco (UFRPE) – Recife – PE – Brazil

*raquel.silvas@ufrpe.br

ABSTRACT

The variability of sea levels is a natural phenomenon that has been aggravated by global warming, which can cause social and economic damage. The complexity of these aspects can be understood through the study of autonomous cells, which in turn offer the possibility of a simple representation for complex and dynamic systems. Applying this methodology makes it possible to study this system in a deterministic scenario or one that can be influenced by external variables. The results obtained in this study extend to a new perspective for modeling environmental phenomena as a way of exploring their dynamics exposed to their own characteristics and the influence of factors external to their nature.

Introduction

Natural events are often modeled using mathematical equations. However, some aspects are too complex to be adapted to such a form, requiring their representation to be based on concepts of complex systems [1]. These systems, which range from financial markets and disease propagation to ecosystems, have internal and external interactive agents, and this relationship is necessary to understand their modeling.

Complexity is understood from the point of view that, in a collection of individual agents, they have autonomy of action and unpredictability, but have an influence on the context of other agents, in other words, they exhibit non-linear behavior. Despite this property, the elements present in complex dynamic systems presuppose that local rules trigger global rules for the composition of this organization.

An example of a complex system is rising sea levels, which make physical and social aspects of civilizations vulnerable. This vulnerability is based on the definition adopted by the Intergovernmental Panel on Climate Change (IPCC): “vulnerability is the degree to which a system is likely or unable to cope with the adverse effects of climate change, including extreme climate variability” [2]. This environmental issue is currently causing notorious problems in terms of urban mobility, but it is estimated that, in the future, these problems will be exacerbated to the point of causing flooding in coastal urban areas [3].

Despite studies, the situation is not treated as urgent, even though there are simulations and sufficient information on the new dynamics that coastal cities will have to experience in the future if there is no change in approaches to respond to warnings about water damage [4].

The aim of this work is to develop a computer model based on cellular automata in order to simulate changes in sea level based on its variability according to internal and external factors - such as the increase of global temperature. Through the analysis, this paper seeks to promote discussions around planning policies and the dissemination of information that will serve as an incentive for further studies using cellular automata and tools aligned with the approach adopted.

Methods

The approaches adopted for the development of this study diverge in the consideration of an external agent as a variable. In this case, the additional variable is the incremental increase in temperature, since this factor drives the rise in sea level and is directly related to global warming and its effects. Initially, not using this conditional allows an analysis of the natural behavior of the cellular automaton without external influence. This makes it possible to observe the system's own dynamics and identify patterns of local sea level rules.

For this purpose, the following scope of implementation is considered: a one-dimensional cellular automaton is used to simulate cells aligned in binary states, which means that 0 is for an inactive cell and 1 is for an active cell. This representation, in the context assumed, is used to indicate a point along the sea coast with or without significant changes in sea level. Modulating the problem from this perspective makes it possible to analyze patterns of propagation of variations over time so that, in this case, local changes in water level can

influence a global coastal system. The construction of this approach consists of structuring the 250 rule, introduced by Stephen Wolfram, which defines how a cell changes state based on the interaction with its immediate neighbors [5].

The model starts with a single active cell in the center of a line made up of 128 cells and evolves over 64 generations, generating dynamic patterns over time between different points on the coastline, where a significant increase can trigger sequential events. However, it is understood that the sea level is not linear since it is hostage to external elements, such as the increase in global temperature, the intensification of the polar ice caps, changes in currents, among others, so that not only is the rate of sea rise affected, but it also introduces greater unpredictability into the behavior of this system. In this sense, the introduction of temperature as an external agent makes it possible to evaluate the impact on the evolution of the system, showing how these factors can modify the initial trend and amplify the observed variability.¹

By applying temperature as a modifying agent in the system, the procedure assumed introduces probabilistic behavior, allowing unexpected changes to be incorporated into the behavior of the water level rise. This approach is closer to real processes, since the deterministic nature is not enough to deal with changing and variable events. The automaton evolves by incorporating temperature variances - taking into account the climate changes recorded each year - into subsequent generations. This means that, as the temperature increases, the chance of unexpected variations also increases, meaning that in a scenario with a higher temperature, the system becomes more susceptible to sudden and less predictable changes.

Results

The comparison between the two approaches suggests differences between considering an ideal and a real scenario. This is due to the unpredictability of the variables linked to sea level rise. In this sense, Figure 1 results in a uniform increase in the sea coast, since, according to the analysis explored, local rules determine global behavior. In short, when the dynamics of the system are not influenced by external factors, such as temperature, the evolution of this organization tends to be stable. Therefore, in the context of sea level rise, this approach suggests that the process of variability occurs in a more linear way, without the abrupt or non-linear fluctuations associated with climatic or thermal events. This characteristic points to the need for real variables so as not to mask the real complexity of the phenomenon.

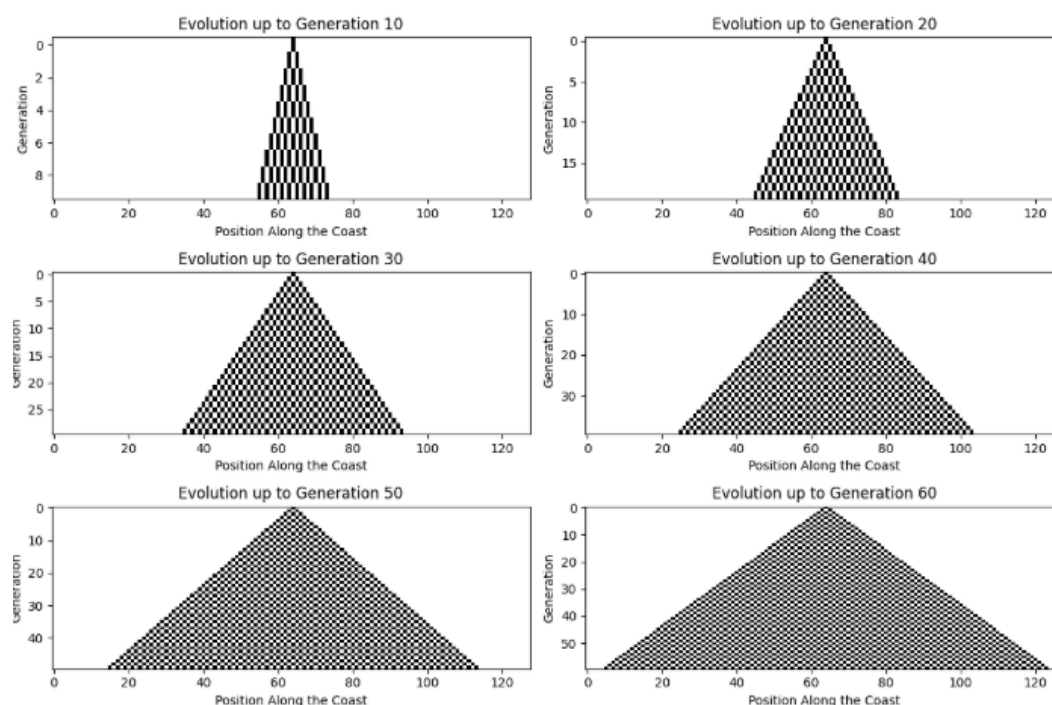


Figure 1. Sea level change without external agents in a One Dimensional Cellular Automata approach.

On the other hand, by adapting this system, assuming the complexity as a determining factor for its monitoring, it is understood that the use of external variables - in this case, temperature - makes its evaluation reliable to the real effects of this phenomenon. The introduction of probabilistic behavior, in other words, the

¹ Both source codes can be found at: <https://github.com/xraquelsilva/CellularAutomataEpidemiology2025>

application of the influence of temperature – which varies annually – is incorporated into the cell, determining a state proportional to this condition, more accurately translating the probability of unmapped variations. Figure 2 shows this lack of linearity, where micro variations can have macro impacts on the assessment of natural phenomena.

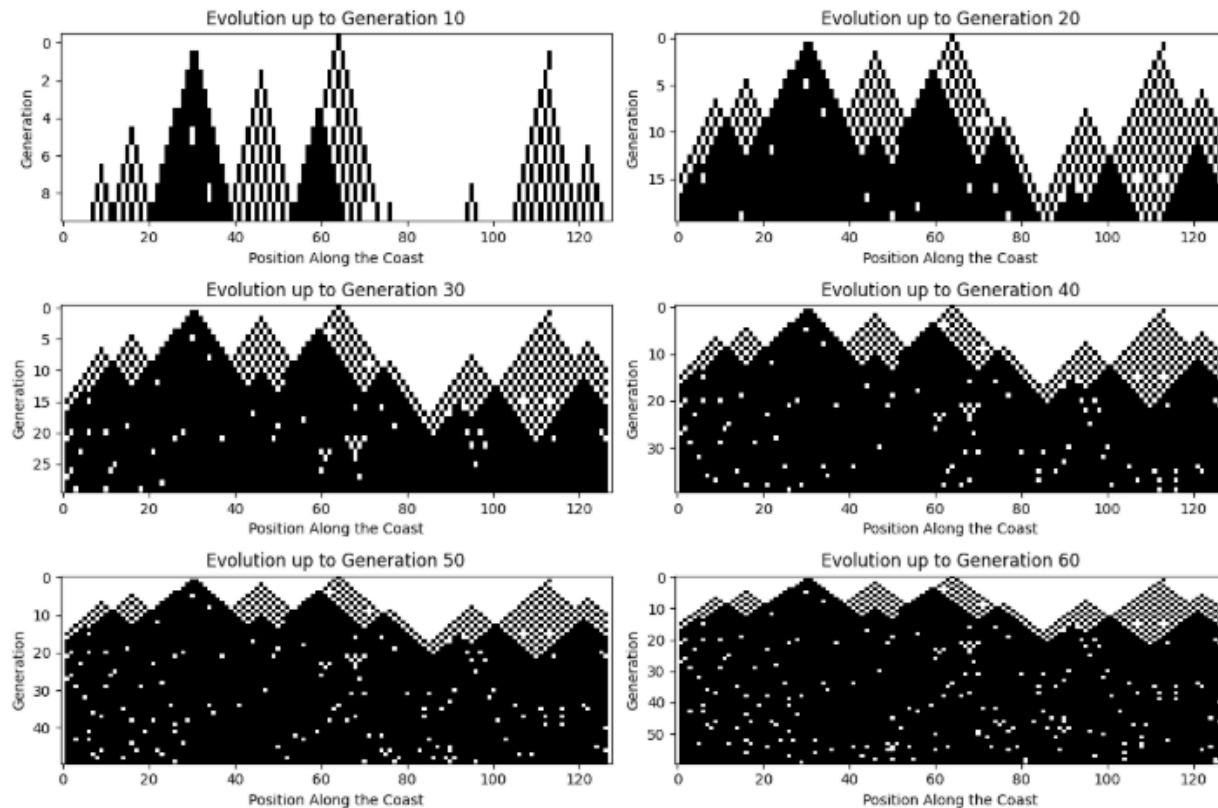


Figure 2. Sea level change with temperature change in a One Dimensional Cellular Automata approach.

The difference in behavior reflects the complexity and dynamism of events in nature. It is understood that, when submitted to conditions specific to their system and to human influence, the processes of biosystems do not follow linear trajectories, but are impacted by multiple interdependent variables. In the case of sea levels, in addition to rising global temperatures, factors such as melting glaciers, variations in ocean currents and changes in precipitation patterns contribute to a non-uniform evolution.

The cellular automata approach makes it possible to investigate emerging patterns and test different scenarios using different conditions, offering a realistic representation. This modeling points to the future implications of climate change, enabling critical thinking to form mitigation and adaptation strategies for vulnerable regions.

Discussions

Sea level, studied using one-dimensional cellular automata, formulates hypotheses about the complex dynamics of its nature in an easy-to-adapt way. Exploring these aspects, based on time-varying local rules, makes it possible to analyze the progression of long-term trends.

Simulating the future states present in this system makes it possible to understand how changes can affect an entire configuration, with impacts felt in different areas. Through the model presented, the effects and risks of current environmental conditions can be understood, guiding strategies to minimize the damage caused by human interference as a way of mitigating harmful impacts on both nature and civilization.

References

1. Pearce, N., Merletti, F. Complexity, simplicity, and epidemiology. *Int J Epidemiol.*, 35(3), 2016. <https://doi.org/10.1093/ije/dyi322>.
2. Organização Meteorológica Mundial (OMM). Relatório do Painel Intergovernamental sobre Mudanças Climáticas (IPCC) - AR6 Mudanças Climáticas 2021: A Base das Ciências Físicas. Released on August 9, 2021. Available at: <https://www.omm.org/pt>. Accessed on: 05 Feb. 2025.

3. Climate Central. Coastal Risk Screening Tool. Available online: <https://coastal.climatecentral.org/>. Accessed on: 05 Feb. 2025.
4. Kulp, S. A.; Strauss, B. H. New Elevation Data Triple Estimates of Global Vulnerability to Sea-level Rise and Coastal Flooding. *Nature Communications*, [S.l.], v. 10, n. 1, p. 4844, 2019. Available at: <http://www.nature.com/articles/s41467-019-12808-z>. Accessed on: 05 Feb. 2025.
5. Wolfram, S. Statistical Mechanics of Cellular Automata. *Rev. Mod. Phys*, 55, 601-644, 1983.